Xiaoshan Wu

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EDUCATION

Harvard University — Cambridge, MA, USA
Visiting Scholar (Embodied AI & 4D World Models) — *Apr. 2026 – Present*
– **Advisors:** Yilun Du and Mengyu Wang.

The University of Hong Kong (HKU) — Hong Kong
Ph.D. in Electrical and Computer Engineering (Focus: 3D/4D Vision, World Models & Embodied AI) — *Sep. 2023 – Aug. 2027*
– **Awards:** Hong Kong PhD Fellowship (**HKPFS, Top Tier**), HKU Presidential Scholarship.
– **Advisor:** Xiaojuan Qi. **GPA:** 4.0/4.3.

Zhejiang University (ZJU) & UIUC — China & USA
Dual B.S. in Computer Engineering (ZJUI Institute) — *Sep. 2019 – Jun. 2023*
– **Honors:** Outstanding Graduate (Rank 5), Dean's List. **GPA:** 3.99/4.0 (ZJU), 3.75/4.0 (UIUC).

RESEARCH INTERESTS

My research connects **3D/4D perception, generative world modeling, and memory for embodied intelligence**. My work spans geometry-grounded robot control, autoregressive video generation, and multi-view spatial reasoning. I am now interested in self-improving physical agents through agent–world model co-evolution.

PUBLICATIONS & PREPRINTS

** Equal contribution; † project lead. Published, preprint, and under-review work.*

First- and Co-first-author Work

1. **Verifiable 4D World–Action Model for Robust Robot Control.**
X. Wu*, R. Cai*, Y. Bao, Z. Su, A. Wu, F. Wu, Z. Li, Z. Qi, H. Chen, W. Ye, J. Zhang, K. Zhou, M. Wang, and Y. Du.
Under review at ICLR 2027. *First author*
S4NDBOX bridges visual imagination and robot control through explicit 4D futures, improving **physical plausibility** (+11.2% relative spatiotemporal IoU vs. RGB-only) and enabling **pre-execution geometric verification** (observed +7.9 pp success within three attempts vs. random ordering). **OOD generalization** over RGB-only: **+13.3 pp** success on RoboTwin Hard (clean-only training); **+40.0** task-score points on single-view **zero-shot sim-to-real** settings.
2. **GaMe: Geometry as Memory for Multi-View Spatial Reasoning.**
X. Wu, X. Lyu, Y. Yu, M. Liu, S. Shi, Z. Wang, and X. Qi.
Under review at ICRA 2027. *First author*
3. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory.**
Y. Yu*, **X. Wu***, X. Hu, et al.
arXiv:2512.04519, 2025. *Co-first author*
4. **EAG3R: Event-Augmented 3D Geometry Estimation for Dynamic and Extreme-Lighting Scenes.**
X. Wu, Y. Yu, X. Lyu, et al.
NeurIPS 2025 · **Spotlight**. *First author*
5. **LiFR-Seg: Anytime High-Frame-Rate Segmentation via Event-Guided Propagation.**
X. Wu, X. Lyu, Y. Yu, B. Wang, Z. Wang, and X. Qi.
ICLR 2026. *First author*
6. **EVSplat: Uncertainty-Aware Gaussian Splatting for Dynamic Scenes with Event Camera.**
Y. Yu*, **X. Wu*†**, X. Lyu, Z. Wang, and X. Qi.
Under review at AAAI 2027. *Co-first author; project lead*
7. **LiFR v2: Completion-Augmented Event Propagation for High-Rate Dense Prediction.**
T. Wan*, **X. Wu*†**, Y. Yu, et al.
Under review at IEEE TPAMI; arXiv:2609.25803, 2026. *Co-first author; project lead*
8. **Event-Based Depth Prediction With Deep Spiking Neural Network.**
X. Wu, W. He, M. Yao, et al.
IEEE TCDS, 16(6):2008–2018, 2024. *First author*

Collaborative Work

9. **Efficient and Accurate Neural-Field Reconstruction Using Resistive Memory.**
Y. Yu, X. Zhang, S. Wang, . . . , **X. Wu**, et al. **Nature**, 654:642–651, 2026.
10. **FoundationGeo: Learning Spatial Pixel-Wise Fields for Monocular Metric Geometry.**
M. Liu, X. Lyu, T. Ren, P. Dai, **X. Wu**, et al. **ECCV 2026**.
11. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization.**
X. Lyu, M. Liu, **X. Wu**, et al. **CVPR 2026**, pp. 7577–7587.
12. **OptiGeo: Efficient Monocular Geometry for Embodied Perception in Optically Challenging Scenes.**
M. Liu, T. Liu, J. Xia, X. Lyu, **X. Wu**, et al. **CoRL 2026**.
13. **Continuous-Time Digital Twin with Analog Memristive Neural Ordinary Differential Equation Solver.**
H. Chen, J. Yang, J. Chen, . . . , **X. Wu**, et al. **Science Advances**, 11(22):eadr7571, 2025.
14. **Topology Optimization of Random Memristors for Input-Aware Dynamic SNN.**
B. Wang, X. Zhang, S. Wang, . . . , **X. Wu**, et al. **Science Advances**, 11(16):eads5340, 2025.
15. **Charge-Selective 2D Heterointerface-Driven Multifunctional Floating Gate Memory for In Situ Sensing-Memory-Computing.**
C. Li, X. Chen, Z. Zhang, **X. Wu**, et al. **Nano Letters**, 24(47):15025–15034, 2024.

TECHNICAL SKILLS

PyTorch, CUDA, Diffusers, Open3D, OpenCV.

SELECTED AWARDS

Hong Kong PhD Fellowship (HKPFS, 2023); HKU Presidential Scholarship (2023).
Outstanding Graduate of Zhejiang University (2023); NeurIPS (2025) and ICLR (2026) Travel Awards.